

Louis Malenfant-Poulin . Jarvis Consulting

I am Louis Malenfant-Poulin, a recent graduate in computer science from Université de Montréal. My experience in competitions such as the ICPC led me to learn much more about algorithms and optimization. This competition also allowed me to practice teamwork under pressure, as solving multiple very hard programming problems under a very short amount of time with two other people required a lot of cooperation. I also led a cybersecurity club where I learned to teach and to lead others in competitions in a domain where understanding is always more important than knowing. Finally, I always participate in computer science events like the advent of code, where I can keep practicing my abilities in Python, C, C++ and many more languages.

Skills

Proficient: Python, Java, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git, Pandas/Numpy

Competent: C/C++, React, Haskell, Docker, Machine learning

Familiar: Burpsuit, Wireshark, Pentesting tools, Javascript, Flask

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_LouisMalenfant-Poulin

Cluster Monitor [GitHub]: Implemented a Cluster monitoring program that allows the user to see information on their machine's statistics and stores them. This project also allows the user to obtain information on their computer's usage, such as memory free, I/O and more by setting up a cronjob that looks it up every minute. This project used shell commands through shell files that called for the execution of information gathering functions such as lscpu and stored them in a sql database.

Highlighted Projects

Complex raytracer: Created a functioning raytracer that rendered images with both precision and speed. This raytracer was implemented in C++ and used OpenGL. It implemented various types of collisions and refractions with a variety of surfaces. Optimization was done through AABBs that helped speed up the raytracing by minimising the checks. This raytracer was also able to simulate collisions with different types of surfaces such as liquids and smoke.

Machine learning project: Developed a machine learning model that could take as input information about a building and predict climatisation usage throughout. This project involved the use of lstm as the information was sorted through time and the loss of order would have rendered the data unreliable. This project also involved gathering and treating large datasets such as normalizing some datapoints and transforming times so that they would be perceived as circular by the program.

Professional Experiences

Jarvis data engineer, Jarvis (2025): Worked in an Agile environnement to implement various data gathering tools and to develop my skills in SQL, Shell and more. Implemented a dockerized version of a computer system analytics tools that gathered information continuously and automatically.

Full-stack internship, SOGI (2022): Developed an employee management system for large enterprises through the use of technologies such as React and SQL. Made a responsive interface for managers to interface with. Created database interfacing applets to quickly and easily enter data.

Education

Université de Montréal (2022-2025), Bachelor in computer science, Computer science and operational research

Miscellaneous

- Cybersecurity club leader
- Supported students through academic difficulties
- Participated in Northsec, ICPC, CSGames
- Part of the administration council of my bachelor's graduate students